

NIKOLAY DEINEGO

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, C, Java, C#, SQL, x86-64 Assembly

Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, ROS 2 Humble, Prisma, OpenCV, MoveIt

Databases & Infrastructure: PostgreSQL, SQLite, Docker, Linux, RESTful APIs, WebSockets

Tools: Git, Wireshark, PCAP, Foxglove Studio, pytest, JUnit

EXPERIENCE

Software Lead (Co-Founder)

Jun 2026 – Present

Otis Dynamics | C++, ROS 2, Docker, MoveIt

Burnaby, BC

- Architected a containerized ROS 2 Humble middleware stack using Docker and C++ (`rc1cpp`), standardizing the `ament_cmake` build system across the team to streamline node development.
- Engineered the robot's foundational kinematic model (`humanoid_description`) using URDF and 3D meshes, establishing the structural baseline for upcoming manipulation and hardware-control layers.
- Built a real-time visualization pipeline utilizing `robot_state_publisher` and the Foxglove WebSocket bridge, enabling live state rendering and structural debugging prior to physical hardware deployment.
- Integrated core orchestration libraries (`ros2_control`, MoveIt, Eigen3, OpenCV) into the base image, provisioning the necessary hardware abstraction and computer vision interfaces for future milestones.

TECHNICAL PROJECTS

Maturin | TypeScript, React, Node.js, Express.js, PostgreSQL

May 2026 – Jul 2026

- Engineered a cycle-aware dependency graph using BFS traversal and topological sort, resolving 500-node network validations in under 0.2ms to prevent circular database insertions.
- Architected an enumeration-safe authentication system utilizing `bcrypt(12)` hashing and 256-bit crypto-random tokens, successfully mitigating timing attacks and brute-force vulnerabilities.
- Designed a 3-tier RBAC approval workflow managing 14 relational PostgreSQL models, securing diff-based state mutations across projects, phases, and milestones.
- Implemented request-time derived analytics, optimizing Prisma queries to eliminate N+1 bottlenecks and computing live 3-factor health scores across 500-entity datasets in ~400ms.

NetGuard | Python, AF_PACKET, TCP/IP

Oct 2025 – Dec 2025

- Engineered a custom packet parser using `AF_PACKET` raw sockets and `struct`, bypassing standard parsing libraries to achieve a baseline decoding throughput of ~310,000 packets/second.
- Architected a multi-threaded capture pipeline to isolate the TUI, preventing 0.68ms render blocks from stalling the intake queue and causing OS-level packet drops during high-throughput ingestion.
- Implemented stateful TCP tracking algorithms for SYN-flood and port-scan detection, maintaining a 0% false-positive rate on standard web traffic and isolating 5,000 PPS attacks in under 20ms.

FutureSpend | Python, FastAPI, Next.js, Gemini API

Mar 2026

- Authored strict API contracts for 20 RESTful endpoints serving 8 distinct frontend views, eliminating integration blockers and enabling independent parallel development.
- Implemented server-side graceful degradation via a rule-based fallback engine, ensuring 100% feature availability for the financial coach during LLM rate limits and timeouts.
- Built a scheduled extraction pipeline utilizing the Google Calendar API to process 7-day rolling windows into a deterministic spending forecast, achieving an execution latency of under 5ms.

EDUCATION

Simon Fraser University (SFU)

Expected Dec 2028

Bachelor of Science in Computer Science

Burnaby, BC

- **Relevant Coursework:** Data Structures & Programming, Introduction to Computer Systems, Software Engineering, Introduction to AI, Robot Learning & Decision-Making, Probability & Statistics.